Actual all-node moments and whole-chart precision

14 November 2026; 38 authored Lean statements
Source `3299b929faad599061b1c13eed9de02ef7e806f2`

Construction. For the same completed auxiliary orientation polynomial, the new component constructs first-jet moment factors at every retained nonzero source node, then derives a finite geometric approximation for every coefficient of its actual crossing chart. The approximation is per entry; compact whole-polynomial matrix factors are a separate open obligation. All records are unsigned [pcc:unverified].

Actual inputs. Let $L = 2^a$, $S = \{-L, \dots, L-1\} \subset \mathbb{Q}$, $t \in S \cap \mathbb{Z}$ nonzero with $v_2(t) = i < u \le a$, and actual ports $x, y < L$ with equal bit i. Use natural multiplicities m_j, the same completed $W_{xy}(T)$, and

$$r = x \bmod 2^u, \quad h = x \operatorname{div} 2^u, \quad \delta_x = y - x, \quad \delta_r = y - r, \quad \alpha = 2^{a+1}.$$

The reference c is the existing nonzero diagonal first jet at the negative pole -2^i, with valuation $w = a + 1 - i + G_i$, where $G_i = m_i - \sum_{j<u}(j-i)_+ m_j$. It is not the potentially zero local first jet at t. Define $P_x(Z) = W_{xy}(t + \alpha Z)/c$ and P_r similarly. The previous all-node source derives $P_x(0) = 0$ and normalized coefficient $n+1$ of weight at least n, hence integral whole polynomials [pcc:unverified].

First-jet moments. Literal prototype denominators give even slopes

$$e_t = \frac{2^u}{t - \delta_r}, \qquad e_j = \frac{2^u}{2^j - \delta_r}, \qquad D_H(H) = (1 + e_t H) \prod_{j \in M_{xy}} (1 + e_j H)^{m_j}.$$

Its constant is one and coefficient b has valuation at least b. The actual normalized first-jet family is its prototype first jet times $D_H(h)^{-1}$. Standard inverse-series truncation below k yields constructed $2^u k$ factors, left weight b, integral right, and normalized residual k. Actual c scaling gives left weight $w + b$ and residual $w + k$. No positive-multiplicity or nonzero local-jet assumption enters this profile [pcc:unverified].

At a positive retained anchor with $m_i > 0$, every actual first crossing jet and corresponding right moment factor is zero. This control concerns the first jet only [pcc:unverified].

Actual polynomial denominator and geometric error

Derive the identity from W. Exact retained-mask comparison and the original nodal puncture identity give

$$(T-\delta_x)W_{xy}(T) = q(T-\delta_r)W_{ry}(T), \qquad q = \prod_{j\in M_{xy}} \left(\frac{2^j-\delta_r}{2^j-\delta_x}\right)^{m_j}.$$

Only actual nonzero denominators are inverted. No local first jet is cancelled. Put $d_x = t-\delta_x$, $d_r = t-\delta_r$ and $\lambda = \alpha/d_r$. Construct

$$D(Z) = q^{-1}\left(\frac{d_x}{d_r} + \lambda Z\right), \qquad N(Z) = (1+\lambda Z)P_r(Z).$$

The literal row identity proves $DP_x = N$. Actual crossing-bit denominators give $q^{-1} \equiv d_x/d_r \equiv 1 \pmod 2$ and $v_2(\lambda) \geq 1$. Thus $D-1$ has content at least one and P_x, N are integral. These facts are proved for the actual inputs, not supplied as favorable premises [pcc:unverified].

Finite construction. Define

$$S_k(D) = \sum_{n=0}^{k-1}(1-D)^n, \qquad A_{x,k} = NS_k(D).$$

The finite geometric identity is $DS_k = 1-(1-D)^k$. If $D-1$ has content one, multiplication by D preserves a desired finite content bound on an integral polynomial: starting at content zero, repeatedly write $R = DR-(D-1)R$ to increase the bound to k. This yields

$$\mathrm{Content}_k(P_x - A_{x,k}), \qquad \mathrm{Content}_{w+k}\big(W_{xy}(t+\alpha Z) - cA_{x,k}(Z)\big).$$

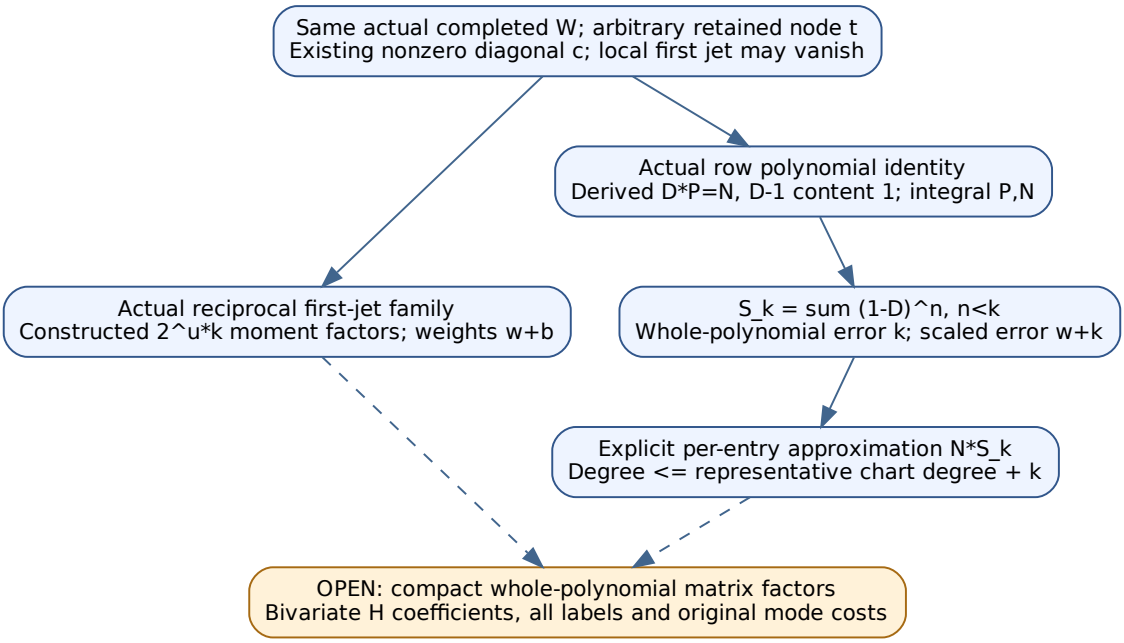
Here content means the bound holds for every polynomial coefficient, including all higher chart coefficients. It permits a vanishing local first jet and uses no inverse of it [pcc:unverified].

Degree accounting. The actual D is affine in Z. For $k>0$, the proof bounds $\operatorname{natdeg} S_k \leq k-1$, and

$$\operatorname{natdeg} A_{x,k} \leq \operatorname{natdeg} P_r + k.$$

The $k=0$ approximation is zero and is handled separately. This bounds polynomial degree, not coefficient bit sizes, construction time or storage [pcc:unverified].

Two constructions and the compact-factor gap



The $2^u k$ matrix factors concern the *first jet*. The full-polynomial approximation $A_{x,k}$ still depends on the individual row through q and d_x. No compact full-polynomial matrix factorization is claimed by this component [pcc:unverified].

Next discriminating construction. Write the actual row quotient as H, construct the polynomial denominator

$$\mathcal{D}(H, Z) = (1 + e_t H + \lambda Z) \prod_{j \in M_{ry}} (1 + e_j H)^{m_j},$$

and its finite geometric approximation. Prove coefficient-H^b content at least b, then truncate $b \geq k$ with an actual error bound at integer h. This is a prospective route to compact whole-polynomial factors; its coefficient bridge and matrix product are not part of these 38 statements [pcc:unverified].

Every residue/moment label must be charged. The original primary interface allows at most $6M/D_i$ modes per weight, $D_i = L/2^i$; a 2^u inventory can exceed its old 2^{i+1} tile count. Full coefficient construction, entry and bit costs, degree, precision, recovery, recursion, public relations, finder, field and scalar extraction, retries and expected cost remain explicit obligations.

Source bindings, validation and preserved failures

The new isolated project builds all 38 authored statements without warnings. The pinned shared gate audits 39 declarations, zero extra trusted axioms, exit 0. Import-inclusive fresh replay is running, not passed. Six central printed dependencies contain only `propext`, `Classical.choice`, `Quot.sound`. All source records are unsigned [pcc:unverified].

`actual_node_moment_profile`	PCC-20261114-ae019d
`actual_row_denominator_identity`	PCC-20261114-b2de93
`actual_whole_chart_precision`	PCC-20261114-1802a0
`actual_weighted_whole_chart_precision`	PCC-20261114-96a569
`actual_approximation_degree`	PCC-20261114-7649ed

Full statements govern their hypotheses. Statement bindings, source pins and the prospective contract are in

`../evidence/claim-pins.json`
`../evidence/source-freeze.json`
`../PROTOCOL.md`.

Gate data is PCC-20261114-03eb52. Local draft007 data is PCC-20261114-496991; that observation alone is not independent gate/replay coverage [pcc:unverified].

Reproducibility. Lean4.35.0, CryptoLean and Mathlib revisions:

`148f5dc389b341401e06d76abe7783c4ccf42f2e`
`cfc55fc58714a59dc06e6e79edc6d1d95140d3c2`.

The source is an exact copy of accepted draft007. All earlier failed sources, eight build logs/exits and local data receipts remain under `../../nodes/moment-evidence/`. Failures include private-import visibility, wrong Lake working directory, an unreduced class port, a deprecated API, and two exact argument/wrapper mismatches. No linter was disabled or desired content/factor witness added.

Sources checked. KB unavailable as journaled. Read exact parent Node/Moment/Content/Anchor denominator, inverse-series, graded entry and row-polynomial proofs; pinned Mathlib geometric-sum and polynomial-degree APIs; earlier primary D.4.3–7 and contraction precision/cost reviews. Standard finite geometric sums and inverse-series moments are reused without a novelty, priority, fastest-method or measured speed assertion.

No native crypto, attack, F4/msolve, timing, scaling, paid or cloud experiment ran. The original norm fixture remains unlaunched under its unchanged checkpoint transport blocker. No measured result updates canonical IC graphs/leaderboards; the diagram records proof dependencies.

The parent arbitrary-node42/44/fresh0 project and all90 artifact pins remain exact, as do earlier parent checkpoints. The protected concurrent PCC histories remain unresolved; auto-merge stays off. No claim rewrite, rechain, side selection, signature, prefix waiver or force push is attempted.